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CNT: An image transform based on complex networks for enhanced texture characterization

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ABSTRACT

This paper proposes the Complex Network Image Transform (CNT) to enhance texture characterization and classification. Our approach leverages the capability of complex networks to effectively represent and analyze intricate phenomena and patterns by modeling the image as a complex network that encodes the detailed relationships among pixels. The topological properties of the resulting network are then employed to generate a set of images that emphasize complex texture patterns, capturing both local and global characteristics through a predefined modeling parameter. The key insight is that these transformed images can serve as enriched inputs for texture analysis and classification tasks. Thus, we evaluate CNT in two distinct scenarios: (i) as a preprocessing step to enhance handcrafted texture descriptors, and (ii) as a data augmentation strategy to improve the training of deep convolutional neural networks. Experiments conducted on five texture datasets show that CNT-generated images are rich in discriminative information. Classification results consistently outperform several baselines, indicating that CNT enhances feature extraction and boosts model generalization. These findings indicate the potential of CNT as a useful tool for texture analysis, and suggest that network-based representations may be worth exploring in related image analysis tasks.

1. Introduction

Representing and characterizing textures is a fundamental task in computer vision, as textures convey visual and structural patterns that describe intrinsic properties of surfaces, such as roughness, smoothness, coarseness, and fineness. Accurate texture characterization therefore plays an important role in a wide range of applications, including medical diagnosis [1], agriculture and remote sensing [2], and advanced material classification [3].

A broad range of texture-analysis techniques has been developed, from classical to learning-based approaches [4,5]. Classical descriptors offer interpretable, specialist-designed representations, but their effectiveness depends on hand-crafted design choices that often struggle in complex real-world scenarios [6]. Deep neural networks, in turn, offer automated feature extraction and have driven substantial gains in texture classification [7–9], but their performance is often constrained by the availability of large and diverse datasets, a common limitation in texture-related problems. Moreover, capturing multiscale properties is a crucial aspect to texture characterization [10]: local methods describe fine-grained patterns such as edges and micro-textures, while global descriptors

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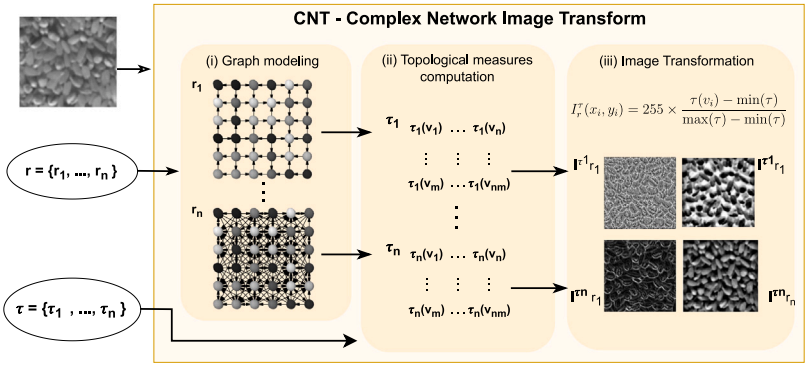


Fig. 1. Illustration of the Complex Network Image Transform (CNT), composed of three stages: graph modeling of pixel relationships, computation of topological network measures, and projection of these measures into new images that provide complementary representations of the original image.

summarize dominant patterns across the whole image, and combining these complementary perspectives has consistently improved performance in applications [11,12]. Thus, developing multiscale and efficient representations that can mitigate data requirements represents an important research direction in texture analysis.

Complex Network (CN) theory offers an alternative approach to texture characterization, since modeling relationships among pixels or image regions encodes structural organization, critical to how textures are perceived [13,14]. This direction was explored by Chalumeau et al. [15] and has since been extended through different network models and characterization strategies [10,16–19], where network properties are used to derive texture descriptors. Importantly, varying modeling parameters, such as edge thresholds and connection radius, enables CNs to capture pixel relationships at different spatial scales, which is particularly relevant to the multiscale nature of textures [10,17]. Another approach proposed by Wang et al. [20] employs CNs as an image preprocessing step for remote sensing classification: images are modeled as complex networks and normalized node degrees are used as inputs to CNNs, demonstrating that network topology can construct image representations beyond conventional descriptor extraction. In this context, we further explore complex networks for generating multiscale representations that can enhance texture classification pipelines.

In this work, we introduce the Complex Network Image Transform (CNT), a new technique that leverages complex network theory to represent and generate image representations, each emphasizing different structural characteristics of the texture. CNT is assessed in two complementary scenarios: (i) as an image preprocessing module that enhances handcrafted texture descriptors and (ii) as a data-augmentation strategy that improves the training and generalization of deep convolutional neural networks (CNNs). Experiments on five publicly available texture benchmarks show that CNT can enhance the discriminative power of handcrafted features and improve the performance of CNN-based models. Thus, the main contributions of this study are:

- A new image-transformation method that highlights both local and global texture characteristics through complex-network modeling;
- A dual evaluation protocol showing CNT's effectiveness for both classical descriptors and deep-learning pipelines;
- An interpretable method that links network-topology metrics to visual texture attributes, providing insights into the structural organization of texture patterns.

2. Complex network image transform

This section introduces the proposed method for enhanced texture characterization based on complex networks. The approach transforms an input image into structurally informed image representations that capture texture patterns from different structural and topological perspectives. These transformed views are explored in two complementary directions: enhancing traditional feature extraction through network-based preprocessed image representations (Section 2.1), and augmenting datasets for deep learning models with semantically consistent transformations (Section 2.2) (see Fig. 1).

Given a grayscale image I with intensities $I(i) \in [0, L]$, a connectivity radius $r \in \mathbb{N}$, and a vertex-level network measure τ, CNT produces a transformed image I_r^τ through the following steps.

Graph construction. Following the image-to-network formulation of [18], each pixel $i = (x_i, y_i)$ is mapped to a vertex v_i and two vertices are linked by a *directed* edge whenever they fall within the same r-radius neighborhood, following the local intensity gradient.

$$E = \left\{ e_{v_i,v_j} \;\middle|\; v_i, v_j \in V,\;\; d(v_i, v_j) \le r^2,\; I(i) < I(j) \right\}, \qquad d(v_i, v_j) = (x_i - x_j)^2 + (y_i - y_j)^2 \tag{1}$$

The orientation of edges makes this construction informative for texture representation. An undirected formulation would link all pixels within the neighborhood symmetrically, so that every vertex initially presents a similar number of connections, yielding

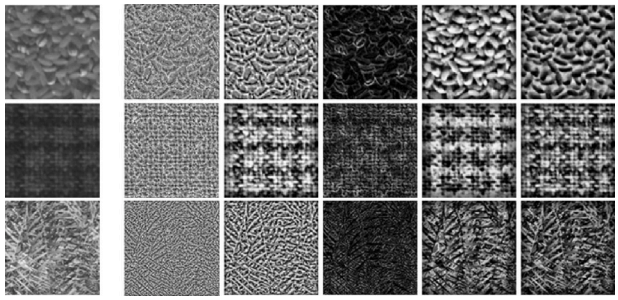


Fig. 2. Examples of the proposed CNT. For each original texture, five transformed images are shown using different network measures and modeling radii.

a regular graph. Prior undirected formulations introduce an intensity threshold to break this symmetry [10,13,16,17], transforming the regular graph into a complex network that has relevant properties for texture analysis [10,13] at the cost of an additional hyperparameter. Orienting each edge by the local intensity gradient achieves the same effect directly: the resulting vertex measures vary from pixel to pixel under the radius alone, reflecting the local intensity variation that defines the texture, without requiring any threshold.

Edge weighting. Each edge e_{v_i,v_j} is weighted by a combination of normalized spatial distance and intensity difference:

$$w(e_{v_i,v_j}) = \begin{cases} \dfrac{|I(i) - I(j)|}{L}, & r = 1, \\ \dfrac{1}{2}\left[\dfrac{d(v_i,v_j) - 1}{r^2 - 1} + \dfrac{|I(i) - I(j)|}{L}\right], & r > 1. \end{cases} \tag{2}$$

Topological measures. For each vertex v_i, four topological measures are computed:

$$k_{\text{in}}(v_i) = \textstyle\sum_{v_j \in V} [e_{v_j,v_i} \in E], \qquad k_{\text{out}}(v_i) = \textstyle\sum_{v_j \in V} [e_{v_i,v_j} \in E], \tag{3}$$

$$f_{\text{in}}(v_i) = \textstyle\sum_{v_j \in V} w(e_{v_j,v_i})[e_{v_j,v_i} \in E], \qquad f_{\text{out}}(v_i) = \textstyle\sum_{v_j \in V} w(e_{v_i,v_j})[e_{v_i,v_j} \in E]. \tag{4}$$

Here, $k_{\text{in}}(v_i)$ and $k_{\text{out}}(v_i)$ represent the network in a binary form, counting the number of darker and brighter neighbors, respectively, within radius r. In contrast, $f_{\text{in}}(v_i)$ and $f_{\text{out}}(v_i)$ consider the weight of the connections, additionally accounting for intensity and spatial differences. Together, these measures capture complementary aspects of the texture structure. Unlike prior work that use these measures directly as feature descriptors [10,13,18], CNT projects each measure back into the image domain to generate a new representation.

Projection. The measure $\tau(v_i) \in \{k_{\text{in}}, k_{\text{out}}, f_{\text{in}}, f_{\text{out}}\}$ is linearly normalized to $[0, L]$ and reassigned to the pixel position of v_i to produce the transformed image.

$$I_r^{\tau}(x_i, y_i) = L \times \frac{\tau(v_i) - \min(\tau)}{\max(\tau) - \min(\tau)}. \tag{5}$$

Each vertex measure is determined by the image-dependent relationships established in the network, aggregating only over the subset of its neighborhood that satisfies an intensity ordering relation. Because of this, the complex network transformation produces a representation that differs from conventional image transformations.

Repeating these steps over a set of radii $\{r_1, \ldots, r_n\}$ and measures $\{\tau_1, \ldots, \tau_m\}$ yields $n \times m$ transformed images per input, each emphasizing structural patterns at a distinct combination of spatial scale and topological property, as shown by the qualitative examples in Fig. 2. Importantly, the transformation preserves the original spatial arrangement of pixels and introduces no external patterns or geometric distortions, so the semantic texture class remains unchanged.

2.1. Improving feature extraction

CNT can be effectively employed as a preprocessing stage to enrich image representations for traditional machine learning pipelines. By generating transformed views, CNT exposes structural information that may not be explicit in the original image, providing additional representations from which texture features can be extracted.

Fig. 3 summarizes the complete feature extraction pipeline: (a) CNT generates from a input image I a set of transformed views I_r^{τ} by varying the radius r and metric τ, as described previously; (b) a texture descriptor d (i.e. Local Binary Patterns, Gray Level Co-Occurrence Matrix) is applied to each view to obtain a set of feature vectors f_r^{τ} that characterize the texture from different topological and spatial perspectives; and (c) these features are progressively fused across network metrics and spatial scales and combined with the features from the original image, creating a unified and discriminative image representation. The fusion is performed in three stages, described below.

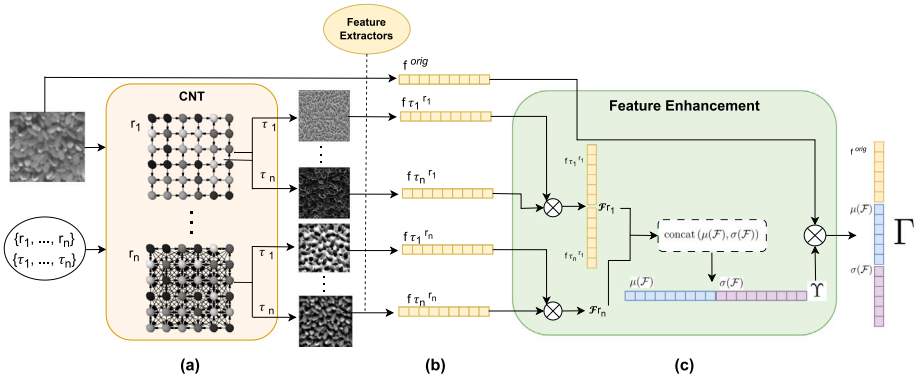


Fig. 3. Overview of the CNT-based feature extraction process. (a) Generation of transformed views I_r^τ using different radii r and network metrics τ; (b) application of a texture descriptor d to extract feature vectors f_r^τ; (c) fusion of multiple vectors into a single representation Γ for classification.

Feature fusion across network metrics. For a fixed connection radius r, the feature vectors f_r^τ obtained from the different network metrics $\tau \in \{k_{\text{in}}, k_{\text{out}}, f_{\text{in}}, f_{\text{out}}\}$ are concatenated to form a composite vector: $\mathcal{F}_r = \bigoplus f_r^\tau$, where $\oplus$ denotes vector concatenation.

Multiscale fusion. Texture patterns often manifest at multiple spatial scales. To capture this, vectors $\mathcal{F}_r$ computed for a set of radii $\{r_1, r_2, \ldots, r_n\}$ are aggregated using point-wise statistical summarization:

$$Y = \text{concat}\big(\mu(\mathcal{F}),\ \sigma(\mathcal{F})\big), \tag{6}$$

where $\mathcal{F} = \{\mathcal{F}_r \mid r \in \{r_1, \ldots, r_n\}\}$, and $\mu(\cdot)$, $\sigma(\cdot)$ represent the element-wise mean and standard deviation computed across the set of vectors, respectively. This produces a unified descriptor Y that summarizes the CNT-derived information across spatial scales.

Integration with original image features. To leverage complementary information from the raw image, the multi-scale CNT-based descriptor Y is concatenated with the feature vector $f^{\text{orig}} = d(I)$ extracted directly from the original image:

$$\Gamma = \text{concat}\big(f^{\text{orig}},\ Y\big). \tag{7}$$

This final vector Γ synthesizes both the original image features and the enriched representations derived from complex network transformations, yielding a robust and discriminative feature set for subsequent analysis, such as classification or retrieval tasks.

2.2. Improving CNN learning

The CNT framework transforms each input image into multiple derived images by applying different graph-based metrics across multiple spatial scales. Since the transformations emphasize different topological properties without introducing artificial noise or irrelevant distortions, and preserving the semantic identity of the original image, they provide complementary views of each texture class and can be used as a data augmentation strategy.

This property is particularly relevant for Deep Convolutional Neural Networks (D-CNNs), which can benefit from diverse training samples to improve generalization. Unlike conventional pixel-level augmentations, such as rotation or translation, CNT-based augmentation operates at a higher abstraction level, manipulating network-derived features rather than raw pixels. Thus, CNT enriches the feature space accessible to the D-CNN, enabling the model to learn from representations that capture the underlying topological structure of textures.

Formally, given an original dataset D containing N images, a set of m network metrics and a set of n connection radii, CNT generates Nmn transformed images. For instance, employing two graph metrics (for instance, in-degree and out-degree) across five radii (for instance, 1 to 5) results in $N \times 2 \times 5 = 10N$ new transformed images. Together with the original N images, this yields a dataset of size $N + 10N = 11N$, representing an eleven-fold increase in data volume.

The full procedure for generating these augmented samples is formalized in Algorithm 1. By integrating graph-structured transformations into the training pipeline, this method can be applied to mitigate overfitting and enhance the model's capacity to generalize across heterogeneous texture instances.

3. Experimental setup

3.1. Datasets

To evaluate the proposed Complex Network Image Transform (CNT), we conducted a comprehensive set of experiments on five established texture databases: CUReT, Outex, USPTex, MBT, and 1200Tex.

Algorithm 1: CNT-based Data Augmentation for D-CNN Training.

Input: Original dataset $\mathcal{D} = \{I_1, \ldots, I_N\}$, graph metrics $\mathcal{T} = \{\tau_1, \ldots, \tau_m\}$, radii set $\mathcal{R} = \{1, \ldots, r_{\max}\}$
Output: Augmented dataset $\mathcal{D}_{\text{aug}}$
Initialize $\mathcal{D}_{\text{aug}} \leftarrow \mathcal{D}$;
foreach *image* $I \in \mathcal{D}$ **do**
  foreach *metric* $\tau \in \mathcal{T}$ **do**
    foreach *radius* $r \in \mathcal{R}$ **do**
      Compute transformed image $I_r^\tau \leftarrow \text{CNT_Transform}(I, r, \tau)$;
      Add I_r^τ to $\mathcal{D}_{\text{aug}}$;
return $\mathcal{D}_{\text{aug}}$;

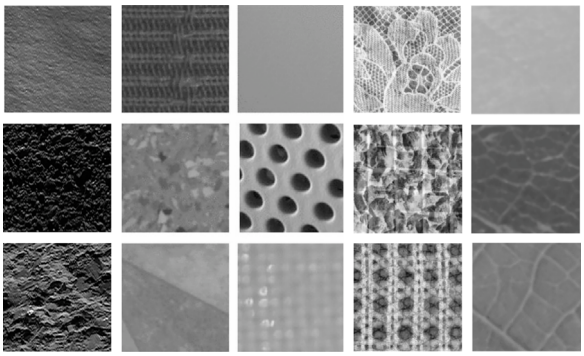


Fig. 4. Texture samples from the five datasets used in this work. Each column represents a different dataset, in order CUReT, Outex, USPTex, MBT and 1200Tex.

- **CUReT**: The Columbia-Utrecht Reflectance and Texture Database (CUReT) [21] contains images of 61 real-world materials covering a wide range of surface types, including specular and diffuse surfaces, isotropic and anisotropic textures, natural and man-made materials. Examples include aluminum foil, artificial grass, concrete, cork, sandpaper, moss, and fur. Each material was captured under different viewing and illumination conditions, resulting in 205 samples of 200 × 200 pixels per class. Following the protocol of [22], we select the subset of 92 samples per class corresponding to viewing angles with a rotation below 60;
- **Outex**: The Outex database [23] provides a standardized collection of surface textures acquired under controlled illumination and rotation conditions, created to be a benchmark dataset for the empirical evaluation of texture analysis algorithms. Following the protocol described in [13], we use the TC_00013 test suite, which contains 68 original images with a resolution of 746 × 538 pixels. Each original image is divided into 20 non-overlapping patches of 128 × 128 pixels, resulting in 1360 texture samples used in the experiments;
- **USPTex**: This natural image dataset is composed by samples found day-to-day (e.g., rice, road scenes, vegetation, walls, clouds, etc.), exhibiting irregular, random, or non-periodic patterns [24]. It comprises 2292 texture samples representing 191 texture classes, with 12 images per class and a fixed resolution of 128 × 128 pixels;
- **MBT**: The Multi-band Texture (MBT) dataset [25] was designed so that images do not have discriminative spectral information, aiming to provide images that allow validation of texture analysis methods based only on texture information. It contains 2464 color texture images (160 × 160 pixels) across 154 classes, each with 16 samples;
- **1200Tex** [26]: A collection of scanned leaf surface images from Brazilian plant species. This dataset includes 20 species, each with 20 exemplars, all resized to 128 × 128 resolution. The collection offers complex texture patterns associated with biological surfaces (see Fig. 4).

3.2. Handcrafted feature extraction

To assess the ability of CNT to enhance conventional texture representations, a diverse set of handcrafted texture descriptors was evaluated. For each descriptor, the baseline corresponds to features extracted from the original image, compared to the CNT-enhanced representation, described in Section 2.1.

The evaluated methods include statistical approaches such as Gray-Level Co-occurrence Matrix (GLCM) [27] and Gray-Level Difference Method (GLDM) [28]; local pattern-based descriptors including Local Binary Pattern (LBP) [29], Complete Local Binary Pattern (CLBP) [30], Local Configuration Pattern (LCP) [31], Adaptive Hybrid Pattern (AHP) [32], Local Grouped Order Pattern

and Non-Local Binary Pattern (LGONBP) [33] and Local Directional Difference and Relational Descriptor (LDDRD) [34]; spectral methods such as Fourier descriptors [35], Gabor filters [36], and the Dual-Tree Complex Wavelet Transform (DTCWT) [37]; as well as the Randomized Encoding Ensemble (REE) [38]. Together, these methods cover a broad spectrum of local, global, statistical, spectral, and hybrid feature extraction strategies.

To evaluate the discriminative power of the extracted features, the Linear Discriminant Analysis (LDA) was applied as the classifier. LDA was selected due to its simplicity and low model capacity, reducing the influence of classifier complexity and allowing the evaluation to focus on the discriminative quality of the extracted representations. A K-fold cross-validation scheme with $K = 10$ was adopted. In each fold, one subset of size N/K (where N is the total number of samples) was used for testing, and the remaining subsets were used for training. This process was repeated ten times, with the average accuracy reported over all folds.

3.3. CNT-based data augmentation

To evaluate CNT as a data augmentation strategy for deep learning-based texture classification, five D-CNN architectures were considered: InceptionV3, ResNet50, ResNet101, DenseNet121, and ConvNeXt-Tiny. Each model was trained under two conditions: (i) using only the original images, and (ii) using an augmented dataset obtained through CNT-based transformations.

The augmentation procedure is controlled by two parameters: the maximum connection radius and the set of network metrics used to generate the transformed images. Four metrics were considered in all experiments: in-degree, out-degree, in-force, and out-force. The maximum radius was varied to control the number of transformed samples generated from each original image. For instance, when five radii and four metrics are used, each original image produces 20 CNT-transformed images, increasing the training set size by a factor of twenty-one.

To prevent data leakage, the datasets were randomly partitioned into training (70%), validation (15%), and test (15%) subsets before applying any CNT transformations. CNT augmentation was performed exclusively on the training subset, so that all transformed images derive exclusively from training samples and no CNT-transformed version of a validation or test image is ever seen during training.

All D-CNNs were trained from scratch for up to 100 epochs (batch size 64); the final model was chosen based on the lowest validation loss. Optimization was performed using the Adam optimizer with a fixed learning rate of 10^{-4}, minimizing the categorical cross-entropy loss. All experiments were repeated three times with different random seeds to ensure statistical robustness, and average performance metrics were reported.

3.4. Implementation details

All experiments were conducted on a server running Ubuntu 22.04.3 LTS, equipped with an Intel Core i9-14900 K processor, 128 GB of RAM and a single NVIDIA RTX 4090 graphics card. The network construction step was implemented in C++ and executed on the CPU, while all deep architectures were trained on the GPU using the PyTorch Image Models library.

4. Results and discussion

This section presents and analyzes the experimental results of the proposed Complex Network-based Image Transformation (CNT) framework. The evaluation covers two main applications: (i) the enrichment of handcrafted texture descriptors and (ii) the use of CNT as a data augmentation strategy for Deep Convolutional Neural Networks (D-CNNs).

4.1. Parameter evaluation for descriptor enrichment

To investigate the influence of CNT parameters on texture representation, we evaluated multiple configurations of the transformation process. In particular, we examined the effects of varying the connection radius and the choice of network-based metrics, considering both isolated and combined configurations, as well as the projection function. Thus, Fig. 5 presents the average classification accuracies obtained across the five datasets and all descriptors using the composite feature vector $\mathcal{F}_r$ concatenated with the original descriptor vector f^{orig}. In this evaluation, the transformation relied on two centrality metrics—*in-strength* and *out-strength*—denoted by the set $\mathcal{T} = \{f_{in}, f_{out}\}$.

The experiment explores various values of r ranging from 1 to 15, selected to capture texture patterns at different spatial scales. As shown in Fig. 5, classification performance generally improves with increasing r, indicating a progressive enhancement in capturing broader contextual and global structural information. This trend stabilizes around $r = 9$, suggesting that beyond this point, the inclusion of additional long-range interactions yields diminishing returns. For instance, in the Outex and USPTex datasets, the most significant accuracy gains are observed between $r = 1$ and $r = 9$, reflecting the increasing ability of CNT to encode discriminative macrostructural patterns. After $r = 9$, performance gains plateau, which indicates that the spatial context captured at this scale is sufficient for most classification tasks within these datasets.

Subsequently, we evaluated the feature vector Γ (Eq. (7)), which integrates features extracted from multiple connection radii. Fig. 6 presents the classification accuracies obtained by combining pairs of radii across the five datasets. The results indicate that the highest classification accuracies are generally achieved when combining radii with substantial scale separation. For example, the pairs $\{01, 09\}$ and $\{01, 15\}$ consistently yield top accuracies in the Outex and USPTex datasets. In contrast, CUReT and 1200Tex achieve their best performances with the radius pair $\{03, 09\}$, although several other combinations produce very similar accuracies.

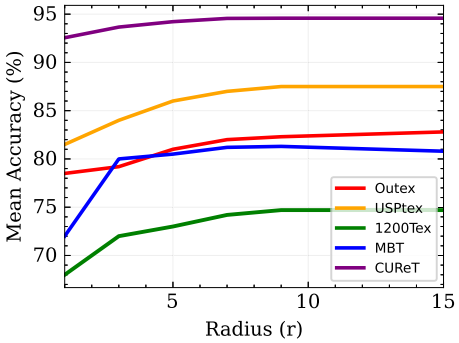


Fig. 5. Accuracies for the feature vector $\mathcal{F}_r$ considering $\mathcal{T} = \{f_{in}, f_{out}\}$ and using different values of r on the four databases.

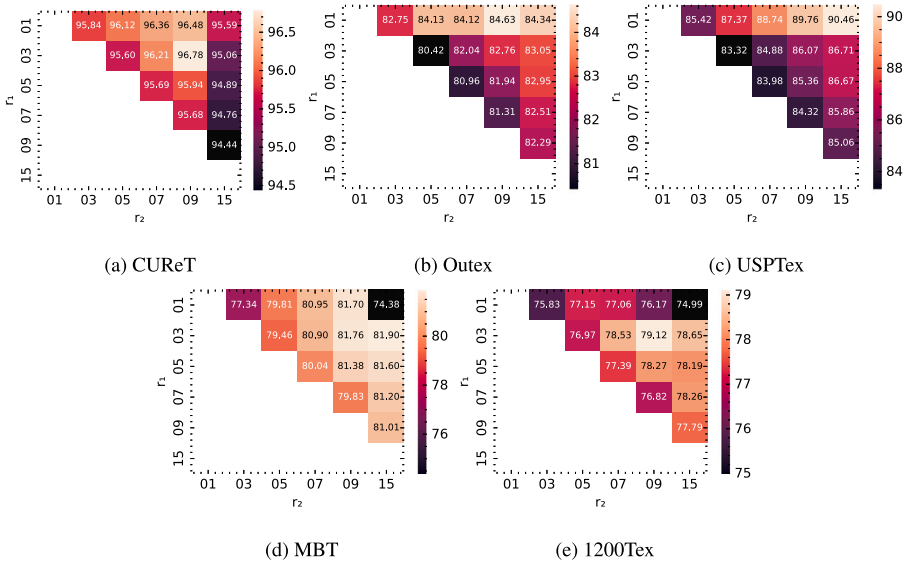


Fig. 6. Accuracies of the feature vector Γ using different values of r and their combinations for the force metric ($\mathcal{T} = \{f_{in}, f_{out}\}$).

Similarly, the MBT dataset achieves its best performance with intermediate radii pairings, such as $\{03, 15\}$ and $\{03, 09\}$, suggesting that these textures are better characterized by larger-scale spatial structures.

Overall, although the optimal scale separation is dataset-dependent, these results reinforce the importance of multiscale modeling in texture analysis: smaller radii capture fine-grained local details, while larger radii emphasize broader spatial relationships and global structures. The combination of these complementary scales enriches the representation and improves its discriminative power. Moreover, the best-performing radius pairs tend to exhibit lower standard deviations, indicating greater consistency among the different feature extractors evaluated in conjunction with the CNT representation. These findings confirm that integrating CNT-derived features across multiple radii improves the model's ability to characterize the diverse structural properties of natural textures.

To further investigate the impact of CNT configurations, we analyzed the influence of the network metrics used in the transformation process. Table 1 presents the classification accuracies obtained by combining degree and strength-based features, while employing the best-performing radius pairs previously identified for each dataset. This evaluation aimed to assess the complementary nature of topological descriptors.

From Table 1, the results indicate that strength-based metrics generally outperform degree-based configurations across most datasets. In CUReT, Outex, USPTex, and MBT, the combination $\{f_{in}, f_{out}\}$ surpasses the degree-only setup $\{k_{in}, k_{out}\}$, suggesting that strength-based metrics provide a richer representation by encoding grayscale intensity variations through weighted edges, whereas degree metrics capture only the number of connections per node.

In contrast, for the CUReT and 1200Tex datasets, the hybrid configuration $\{f_{out}, k_{out}\}$ achieved the highest accuracies, while in Outex and USPTex the second-best results were obtained with $\{f_{in}, k_{in}\}$. This behavior indicates that combining both degree and strength metrics can provide additional discriminative information, as they jointly capture local connectivity patterns and contextual variation. Overall, these findings reinforce the effectiveness of integrating complementary network metrics within the CNT framework, demonstrating that their combination improves the discriminative capability of the resulting texture representation.

Table 1
Mean accuracies of the feature vector Γ using the tuned radius for different combinations of metrics.

Dataset	$\{r_1, r_2\}$	$\{f_{in}, f_{out}\}$	$\{k_{in}, k_{out}\}$	$\{f_{in}, k_{in}\}$	$\{f_{out}, k_{out}\}$
CUReT	{03, 09}	96.78 ± 2.41	96.69 ± 2.51	96.88 ± 2.43	**96.92 ± 2.42**
Outex	{01, 09}	**84.63 ± 4.4**	83.15 ± 4.7	84.29 ± 4.0	84.07 ± 4.3
USPTex	{01, 15}	**90.46 ± 7.8**	82.47 ± 16.3	89.29 ± 10.6	89.15 ± 10.6
MBT	{03, 15}	**81.90 ± 11.9**	81.77 ± 13.0	81.48 ± 11.1	81.69 ± 13.2
1200Tex	{03, 09}	79.12 ± 4.8	80.69 ± 4.7	79.61 ± 5.0	**81.09 ± 5.0**

Table 2
Comparison of the accuracies using only the original image f^{orig} and the Γ representation from the CNT framework for the CUReT, Outex and USPTex datasets. Notation Δ represents the difference in accuracy between f^{orig} and Γ. The best results are shown in **bold** and the second-best in *italics*.

	CUReT			Outex			USPTex		
	f^{orig}	Γ	Δ	f^{orig}	Γ	Δ	f^{orig}	Γ	Δ
GLCM	70.90 ± 1.5	92.92 ± 1.4	+22.02	81.91 ± 2.2	89.85 ± 2.0	+7.94	83.42 ± 2.3	95.37 ± 1.4	+11.95
GLDM	82.07 ± 1.4	95.67 ± 1.1	+13.60	87.13 ± 2.4	89.85 ± 2.0	+2.72	91.67 ± 1.4	95.51 ± 1.2	+3.84
LBP	56.54 ± 1.3	99.61 ± 2.3	+43.07	71.91 ± 2.5	82.65 ± 3.2	+10.74	73.12 ± 2.4	88.26 ± 2.2	+15.14
CLBP	91.05 ± 1.0	92.46 ± 0.9	+1.41	81.03 ± 3.6	81.91 ± 3.0	+0.88	85.08 ± 1.6	81.72 ± 2.5	−3.36
LCP	79.77 ± 1.5	93.76 ± 1.0	+13.99	86.25 ± 2.9	89.12 ± 2.3	+2.87	90.84 ± 1.9	94.98 ± 1.2	+4.14
AHP	89.31 ± 1.0	99.57 ± 0.3	+10.26	87.43 ± 1.7	90.73 ± 2.3	+3.30	94.72 ± 1.1	**97.25** ±1.0	+2.53
LGONBP	97.77 ± 0.7	*99.69 ± 0.3*	+1.92	72.94 ± 0.3	90.51 ± 2.4	+17.57	77.48 ± 2.8	96.80 ± 1.0	+19.32
LDDRD	92.59 ± 0.1	96.97 ± 0.6	+4.38	85.59 ± 2.5	86.93 ± 2.8	+1.34	93.46 ± 1.3	94.42 ± 2.2	+0.96
Fourier	72.08 ± 1.4	92.76 ± 0.9	+20.68	80.51 ± 4.5	85.51 ± 3.0	+5.00	65.97 ± 3.7	88.52 ± 1.6	+22.55
Gabor	81.15 ± 1.5	99.12 ± 0.3	+17.97	81.10 ± 2.7	86.76 ± 2.7	+5.66	87.17 ± 3.0	96.47 ± 1.5	+9.30
DTCWT	81.06 ± 1.3	96.19 ± 0.6	+15.13	85.22 ± 2.3	88.97 ± 2.3	+3.75	90.71 ± 0.9	*97.12* ±1.3	+6.41
REE	95.40 ± 0.4	**99.75 ± 0.2**	+4.35	**92.50 ± 1.9**	*90.88 ± 1.9*	−1.62	97.07 ± 0.7	89.18 ± 1.4	−7.95
Average			+14.07			+5.01			+7.07

Table 3
Comparison of the accuracies using only the original image f^{orig} and the Γ representation from the CNT framework for the MBT and 1200Tex datasets. Notation Δ represents the difference in accuracy between f^{orig} and Γ. The best results are shown in **bold** and the second-best in *italics*.

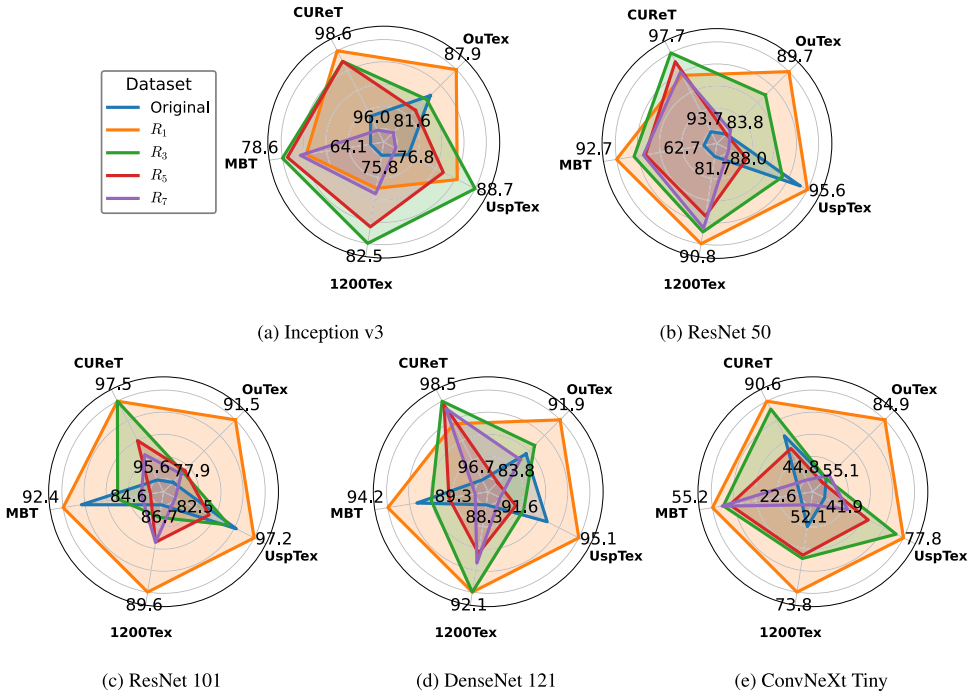
	MBT			1200Tex		
	f^{orig}	Γ	Δ	f^{orig}	Γ	Δ
GLCM	85.71 ± 1.7	86.85 ± 0.0	+1.14	70.50 ± 2.8	84.92 ± 3.9	+14.42
GLDM	92.13 ± 1.7	**95.74 ± 0.7**	+3.61	82.25 ± 2.9	88.25 ± 3.9	+6.00
LBP	65.95 ± 3.5	78.89 ± 2.1	+12.94	68.42 ± 2.6	78.75 ± 3.7	+10.33
CLBP	78.49 ± 2.8	73.58 ± 2.3	−4.91	64.00 ± 1.6	81.75 ± 2.7	+17.75
LCP	83.32 ± 3.2	87.34 ± 2.1	+4.02	77.92 ± 3.9	84.67 ± 3.6	+6.75
AHP	84.38 ± 2.3	89.32 ± 1.7	+4.94	80.67 ± 2.8	81.42 ± 3.0	+0.75
LGONBP	77.52 ± 1.6	87.54 ± 1.9	+10.02	56.50 ± 4.0	*88.75 ± 1.9*	+32.25
LDDRD	85.84 ± 1.6	87.42 ± 2.9	+1.58	77.58 ± 2.5	84.58 ± 3.6	+7.00
Fourier	67.17 ± 4.4	68.59 ± 1.2	+1.42	68.00 ± 3.9	73.00 ± 3.9	+5.00
Gabor	88.96 ± 1.4	91.64 ± 1.0	+2.68	78.34 ± 2.4	85.25 ± 3.9	+6.91
DTCWT	87.87 ± 2.4	91.07 ± 2.4	+3.20	81.34 ± 1.4	86.17 ± 3.8	+4.83
REE	*93.10 ± 0.8*	89.85 ± 1.7	−3.25	88.25 ± 2.2	**89.33 ± 2.9**	+1.08
Average			+3.12			+9.09

4.2. CNT as a framework for descriptor enrichment

To evaluate the enrichment of handcrafted texture descriptors, we compared the performance of several well-established methods in two scenarios: (a) their standalone, traditional application, and (b) their enriched version Γ (Eq. (7)). This comparison quantifies the direct performance gain achieved by augmenting existing descriptors with the rich structural information provided by CNT. The CNT transformations used the strength metric and the optimized sets of radii identified for each dataset: {03, 09} for CUReT, {01, 09} for Outex, {01, 15} for USPTex, {03, 15} for MBT, and {03, 09} for 1200Tex. Tables 2 and 3 summarize the accuracies obtained for both the original descriptors (f^{orig}) and their CNT-enriched counterparts (Γ) across all five texture databases.

From Tables 2 and 3, a clear and consistent trend is that the CNT framework substantially enhances the performance of most texture analysis methods across all evaluated databases. For example, the CUReT dataset achieves the highest average gain (+14.07), followed by 1200Tex (+9.09). This consistent behavior across diverse datasets indicates that CNT improves texture representation and highlights the generalizability of the framework.

In addition to improving overall performance, CNT-enhanced methods also achieve the best accuracies in most datasets. In particular, REE achieves best accuracy in CUReT (95.40% to 99.75%) and 1200Tex (88.25 to 89.33%), AHP in USPTex (94.72 to



(a) Inception v3 (b) ResNet 50

(c) ResNet 101 (d) DenseNet 121 (e) ConvNeXt Tiny

Fig. 7. Accuracies of five D-CNN architectures in five different databases. Each line in the radar chart represents a model's performance on a different CNT augmentation setting (defined by its maximum radius), while the polygon vertices correspond to the accuracy achieved on each of the texture databases.

97.25%) and GLDM in MBT (92.13 to 95.74%). The only exception is Outex, where CNT attains the second-best result. Moreover, reduced standard deviations in several methods (e.g., Fourier, LBP, and GLCM) indicate improved robustness and stability.

The magnitude of the gains is closely related to the complementarity between CNT and each descriptor. For example, methods based on local intensity patterns, such as LBP and LGONBP, show significant improvements (e.g., +43.07 in CUReT and +32.25 in 1200Tex, respectively), as well as frequency-based descriptors such as Fourier and Gabor (e.g., +22.55 in USPTex and +17.97 in CUReT, respectively). In contrast, a few methods, such as CLBP in USPTex and MBT, or REE in Outex, show marginal gains or slight decreases, suggesting a lower degree of complementarity. Overall, these results suggest that CNT introduces complementary structural information derived from the complex network modeling of textures and the projection of vertex measures, rather than simply re-expressing information already captured by conventional texture descriptors.

4.3. CNT as a data augmentation tool

In this section, we present and discuss the results obtained from our experiments, which evaluate the effectiveness of the proposed Complex Network Image Transform (CNT) as a data augmentation tool for deep learning models. Our analysis is based on the performance of five D-CNN architectures across five distinct texture databases. We systematically investigated the impact of the maximum radius parameter, which controls the scale and quantity of augmented data. Specifically, we tested various sets of radii, referred to as R_1, R_3, R_5 and R_7, where the subscript indicates the maximum radius in the set. For instance, R_1 represents the set of radii $\{1\}$, and R_3 represents $\{1, 2, 3\}$.

Fig. 7 presents the results, which are shown as radar charts for a comprehensive view of each model's performance, with one chart dedicated to each D-CNN architecture. This visualization method is particularly effective for simultaneously comparing a model's accuracy across multiple datasets and different augmentation strategies. Each spoke of the radar chart represents a specific dataset, and the performance for each augmentation factor (e.g., R_1, R_3) is plotted as a different line.

The results indicate a significant improvement in performance across all models and datasets when CNT-based data augmentation is applied. This is visually supported by the radar charts in Fig. 7, where the largest polygons, representing the highest accuracies, are consistently associated with the augmented models. This improvement underscores the framework's capacity to enrich the training data with valuable structural information, thereby enabling D-CNNs to learn more robust and generalizable features.

The results also reveal a strong dependence on the maximum connection radius used to generate the augmented images. In most cases, the highest performance gains are obtained with smaller radii (e.g., $R = 1$), indicating that CNT effectively enriches local structural information, which plays a crucial role in texture discrimination. Furthermore, some configurations achieve their best accuracies with R_3, including CUReT (ResNet50 and DenseNet121), MBT (InceptionV3), and 1200Tex (InceptionV3 and

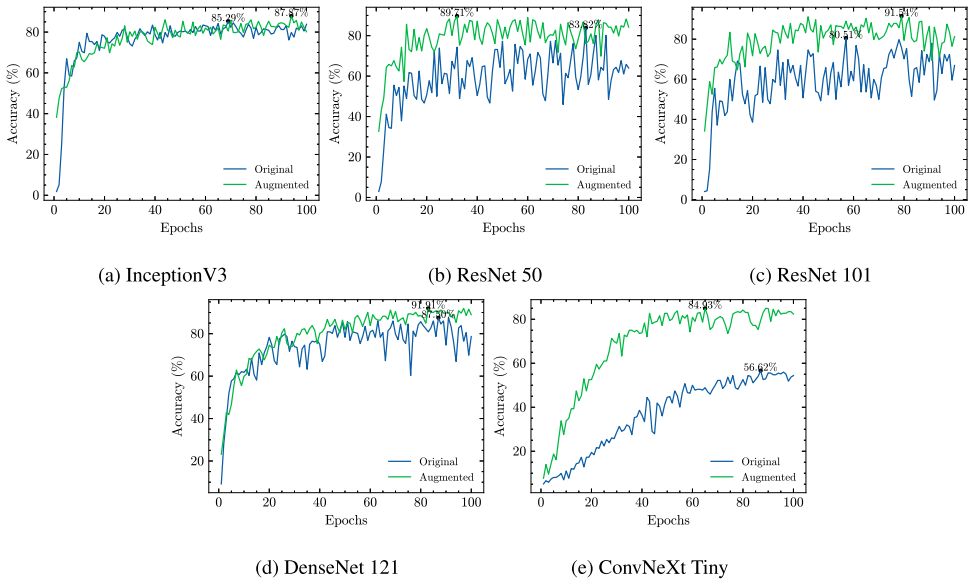


Fig. 8. Validation accuracy curves for five D-CNN architectures on the Outex database. Each graph displays the validation accuracy over epochs for a model trained on the original dataset versus a model trained with CNT augmentation.

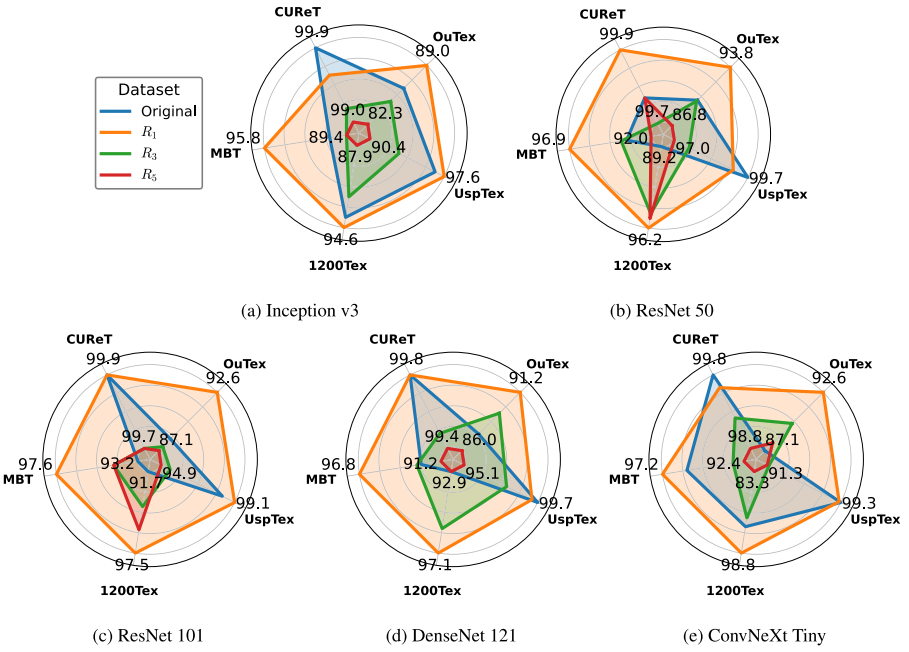
DenseNet121). Since R_3 contains transformed images generated with radii $\{1, 2, 3\}$, these results suggest that combining multiple structural scales can also provide complementary and discriminative information.

Across all experiments, the ResNets and DenseNet consistently demonstrated strong performance, with ResNet50, ResNet101 and DenseNet121 surpassing 89% accuracy in all datasets when augmented with CNT at R_1. Notably, in the USPTex dataset, ResNet101 achieved a peak accuracy of 97.2%. In the remaining datasets, DenseNet121 achieved the highest accuracies, reaching 98.53% in CUReT, 91.91% in Outex, 94.15% in MBT, and 92.08% in 1200Tex, with CNT configurations R_1 and R_3. Furthermore, ConvNeXt Tiny exhibited the most dramatic improvements with CNT; although it did not consistently reach the peak accuracy levels of the other models, its performance increased substantially relative to its baseline. For instance, in Outex, its accuracy increased from 56.98% to 84.92% at R_1, while in MBT it improved from 22.56% to 55.19%. Overall, the consistent improvements observed across all architectures highlight the versatility of the CNT framework as a general-purpose strategy for enhancing texture representations.

The validation accuracy curves in Fig. 8 provide a more granular view of the training dynamics across all architectures on the Outex dataset. Each plot compares a model trained on the original dataset with its counterpart trained using CNT-augmented data R_1. Models trained with CNT-augmented data consistently achieve higher peak accuracies and exhibit more stable learning behavior compared to those trained on the original data. For instance, the original ResNet101 (Fig. 8(c)) shows significant fluctuations and early saturation, whereas its CNT-augmented counterpart presents smoother and more consistent convergence. These results indicate that CNT improves both performance and training stability by providing a more informative learning signal.

To further evaluate the proposed approach, we conducted additional experiments using ImageNet-pretrained models, which were fine-tuned on the texture datasets following the same augmentation protocol. Fig. 9 summarizes the results using radar charts, following the same presentation adopted for the models trained from scratch in Fig. 7. Overall, the use of ImageNet pretraining leads to higher baseline accuracies, and consequently, the improvements obtained with CNT are generally more moderate than those observed when training from scratch. Nevertheless, CNT remains beneficial in several of the evaluated configurations. For example, on Outex, ConvNeXt Tiny improves from 87.13% to 92.64% when fine-tuned with CNT at R_1, compared to 84.92% when trained from scratch under the same augmentation setting (Fig. 7(e)). Similarly, on 1200Tex, ResNet 101 improves from 91.67% to 97.5% at R_1, compared to 89.58% from scratch (Fig. 7(c)). However, the effect of CNT is less pronounced on datasets for which the ImageNet-pretrained models already achieve near-saturated performance. This is particularly evident on CUReT, where the average baseline accuracy is 99.84% and CNT produces only marginal changes: Inception V3 decreases from 99.91% to 99.56%, Resnet50 improves from 99.74% to 99.91%, and ResNet101 and DenseNet121 remain unchanged.

Lastly, CNT is compared to standard data augmentation pipelines, including AutoAugment [39], RandAugment [40], TrivialAugment [41], and Random Erasing [42]. The same experimental protocol was adopted for all methods: augmentation was applied exclusively to the training subset, following rates aligned with CNT configurations R_1, R_3, R_5, generating the same number of training samples. Table 4 reports the classification accuracies obtained on the Outex dataset. Among the conventional augmentation strategies, the best performances are generally achieved by Random Erasing, with DenseNet121 reaching the highest accuracy of 91.17%, followed by ResNet50 (89.33%). Similarly, RandAugment presents competitive results, such as 88.60% with DenseNet121 and 87.86% with ResNet50. Despite these strong results, these methods still underperform compared to the CNT framework, as shown in Fig. 7. These results further indicate that the performance gains achieved by CNT are not solely attributable to the increase in the number of training samples, but also to the structural information introduced by the complex network transformations.



(a) Inception v3 (b) ResNet 50

(c) ResNet 101 (d) DenseNet 121 (e) ConvNeXt Tiny

Fig. 9. Accuracies of five D-CNN architectures in five different databases. Each line in the radar chart represents a model's performance on a different CNT augmentation setting (defined by its maximum radius), while the polygon vertices correspond to the accuracy achieved on each of the texture databases.

Table 4
Comparison of classification accuracy (%) on the Outex dataset using different data augmentation pipelines. Each method is evaluated under three augmentation levels, corresponding to CNT's R_1, R_3, R_5.

	AutoAugment			RandAugment			TrivialAugment			Random erasing		
	R_1	R_3	R_5	R_1	R_3	R_5	R_1	R_3	R_5	R_1	R_3	R_5
Inception V3	83.45	79.77	80.14	86.76	86.76	84.55	81.25	81.98	83.82	86.76	87.86	87.50
Resnet50	84.55	81.25	81.25	44.07	86.76	87.86	83.45	81.61	83.45	87.13	89.33	86.76
Resnet101	83.45	79.77	80.14	86.76	86.76	86.76	81.25	81.98	83.82	86.76	87.86	87.13
Densenet121	83.45	83.45	83.45	86.02	81.98	88.60	88.60	87.50	87.50	91.17	91.17	91.17
ConvNeXt Tiny	65.80	69.48	69.85	70.58	83.82	83.82	69.85	77.20	78.30	71.32	76.47	74.26

4.4. Robustness analysis

This section evaluates the robustness of the proposed Complex Network Transform (CNT) when applied to texture images under different conditions, such as noise, rotation and illumination changes, a desirable property for practical texture analysis applications.

Noise tolerance. To assess noise robustness, salt-and-pepper noise was added to the Outex13 dataset at densities $d \in \{0.01, 0.1, 0.2\}$. For an image containing n pixels, a noise density d affects approximately $d \cdot n$ pixels. The classification performance obtained with the original and CNT-enhanced representations is reported in Table 5. As expected, increasing the noise density generally reduces classification accuracy for both representations. However, the CNT-enhanced representation consistently exhibits a smaller performance degradation across the evaluated descriptors, suggesting that CNT mitigates the impact of noise and provides a more robust texture representation.

Rotation robustness. Rotation robustness was evaluated using the rotated Outex protocol described in [13]. For each texture class, one image was rotated to 11 orientations ranging from 15° to 165°, resulting in 11 samples per class. A leave-one-out protocol was adopted, in which one orientation was used for testing while the remaining orientations were used for training. This setup evaluates the ability of the representations to generalize to an unseen rotation of the same texture.

Illumination robustness. Robustness to illumination changes was evaluated using the Outex TC_00014 protocol. The dataset contains texture images acquired under three simulated illuminants: 2300 K horizon sunlight, 2856 K incandescent CIE A, and 4000 K fluorescent TL84. Following the standard protocol, 680 images acquired under one illumination condition were used for training, while 1360 images acquired under the two remaining illumination conditions were used for testing, directly evaluating the ability of the descriptors to generalize across changes in illumination.

Table 5
Accuracy (%) under noise levels for Outex13. Values in parentheses indicate the variation relative to the original Outex13 accuracy (f^{orig}).

	$d = 0.01$		$d = 0.1$		$d = 0.2$	
	f^{orig}	Γ	f^{orig}	Γ	f^{orig}	Γ
Fourier	62.7 (−17.8)	80.4 (−5.1)	50.8 (−29.7)	76.5 (−9.0)	38.9 (−41.5)	70.1 (−15.4)
GLDM	87.8 (+0.6)	91.4 (+1.5)	85.8 (−1.3)	90.1 (+0.2)	85.7 (−1.5)	88.7 (−1.2)
LBP	71.9 (0.0)	82.7 (+0.1)	69.7 (−2.2)	80.9 (−1.8)	63.2 (−8.7)	79.3 (−3.3)
AHP	88.1 (+0.7)	89.6 (−1.1)	86.2 (−1.2)	87.5 (−3.2)	83.3 (−4.1)	84.9 (−5.8)
LCP	83.8 (−2.4)	86.8 (−2.4)	83.5 (−2.7)	84.8 (−4.3)	77.4 (−8.8)	81.0 (−8.1)
GLCM	80.9 (−1.0)	89.3 (−0.5)	79.9 (−2.0)	87.6 (−2.3)	79.3 (−2.6)	84.6 (−5.2)
CLBP	79.1 (−1.9)	83.4 (+1.5)	72.6 (−8.4)	79.8 (−2.1)	65.7 (−15.3)	72.6 (−9.3)
Gabor	77.1 (−4.0)	85.3 (−1.5)	52.6 (−28.4)	80.5 (−6.2)	42.9 (−38.2)	75.3 (−11.5)
DTCWT	81.2 (−4.0)	87.3 (−1.6)	55.2 (−30.1)	84.5 (−4.7)	42.5 (−42.7)	80.2 (−8.7)
REE	90.9 (−1.6)	90.6 (−0.3)	73.2 (−19.3)	83.9 (−7.0)	54.8 (−37.6)	75.6 (−15.3)

Table 6
Accuracy (%) under rotation and illumination variations on the rotated Outex and Outex14, respectively. Results compare the original descriptors (f^{orig}) with their CNT-enhanced representations (Γ).

	Rotation		Illumination	
	f^{orig}	Γ	f^{orig}	Γ
GLCM	76.20	92.11	57.21	72.57
GLDM	84.09	92.65	67.21	75.29
LBP	78.88	88.10	54.19	66.69
CLBP	64.71	93.98	38.90	70.07
LCP	93.58	96.39	69.71	74.34
AHP	92.78	88.50	67.87	59.34
Fourier	92.37	94.38	55.37	69.78
Gabor	79.28	89.04	59.04	72.21
DTCWT	90.77	94.65	65.44	76.32
REE	96.92	98.93	76.84	78.75

Table 6 summarizes the results for the rotation and illumination protocols. CNT-enhanced representations generally achieve higher accuracy than their corresponding original descriptors in both settings, indicating that the structural representations generated by CNT can provide complementary information that reduces the sensitivity of handcrafted texture descriptors to changes in rotation and illumination.

4.5. Computational complexity analysis

Let a grayscale image of size $H \times W$ be modeled as a directed graph with $N = H \cdot W$ vertices. For a given modeling radius r, each vertex interacts with pixels within its r-neighborhood, which involves $O\left(\frac{(2r+1)^2}{2}\right)$ operations. The graph construction stage requires examining each pixel and its local neighborhood to define directed edges and compute edge weights, resulting in a computational cost of $O\left(N\frac{(2r+1)^2}{2}\right)$. Similarly, the computation of local network-based measures, such as degree- and force-based descriptors, scales linearly with the number of edges and therefore also incurs a cost of $O\left(N\frac{(2r+1)^2}{2}\right)$. The subsequent normalization and projection of the computed measures back to the image domain are linear in the number of pixels, i.e., $O(N)$. Consequently, the overall computational complexity of the proposed transformation is dominated by the modeling and measure computation stages and can be expressed as:

$$O\left(N\frac{(2r+1)^2}{2}\right).$$

Regarding space complexity, the full graph structure is not explicitly stored. Instead, network measures are computed and accumulated across different radii, avoiding redundant storage of edges. The dominant memory cost therefore corresponds to storing network-based measurements for each pixel and each considered radius, resulting in a space complexity of $O(Nr)$. The projection step requires only linear additional memory and does not alter this bound.

To complement the theoretical analysis, empirical runtime measurements of the proposed Complex Network Transform (CNT) were conducted. The execution time was decomposed into two stages: (i) the modeling stage, including neighborhood exploration and edge weight computation, and (ii) the transformation stage, comprising the computation of network-based measures and their projection back to the image domain. Table 7 reports the average runtime per 128 × 128 image, computed over 10 repetitions. As expected, runtime increases with the maximum modeling radius $r_{\max}$, since larger radii induce denser neighborhood structures and a greater number of edges. Nevertheless, it is important to emphasize that the CNT computation is performed in a constructive

Table 7
Average runtime (in milliseconds) of CNT for different maximum radii $r_{\max}$, decomposed into modeling, transformation, and total time per image.

r_{max}	Modeling	Transform	Total
1	1.25 ± 0.14	0.09 ± 0.20	1.33 ± 0.25
5	17.48 ± 1.09	1.19 ± 0.29	18.67 ± 1.13
10	61.44 ± 3.30	3.87 ± 0.81	65.31 ± 3.40
15	125.40 ± 7.62	7.68 ± 0.79	133.08 ± 7.83

manner: when a maximum radius $r_{\max}$ is specified, all intermediate representations for $r = 1, \ldots, r_{\max} - 1$ are obtained implicitly during the same process. Consequently, using a combination of multiple radii (e.g., $r = \{1, 5, 15\}$) does not require recomputing each radius independently, and the effective computational cost corresponds only to the highest radius considered.

5. Conclusion

This paper introduced the Complex Network Image Transform (CNT), a method for texture representation that models images as complex networks and transforms topological information into image representations. By varying the connection radius and selecting different network measures, CNT provides representations that encode intensity relationships and structural variations at different spatial scales. This allows the framework to be used both to enrich handcrafted texture descriptors and to generate additional representations for deep learning models.

When combined with classical texture descriptors, CNT consistently improved classification performance across five benchmark datasets, indicating that the information introduced by the network-based representations can complement existing texture descriptors and enhance their discriminative capability. CNT also showed potential as a data augmentation strategy for deep learning. Models trained with CNT-augmented datasets generally outperformed those trained on the original data and those using standard augmentation methods, including AutoAugment, RandAugment, TrivialAugment, and Random Erasing.

Overall, CNT provides a versatile approach for enhancing texture representations, either through descriptor enrichment or as a data augmentation mechanism for deep learning. The results across the evaluated texture-classification benchmarks support the potential of the proposed representations to capture complementary structural information at different spatial scales and improve classification performance. Future work will investigate the application of CNT to real-world imaging scenarios, including medical, biological, and material imaging, and assess its ability to transfer to image analysis problems beyond texture classification.

CRediT authorship contribution statement

Ana Beatriz S. Zerati: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Luan B. Guerra:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Fabiana R. de Góes:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Leandro Alves Neves:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis. **Lucas C. Ribas:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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